Aryan Gupta

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EDUCATION

transcripts available upon request

University of North Carolina at Charlotte - Charlotte, NC

M.S. in Computer Engineering — *Aug. 2020 – Dec. 2024*

Current Cumulative GPA: 3.830 / 4.000

B.S. in Computer Engineering | *Minor in Software Systems* | *Minor in Mathematics* — *Aug. 2016 – Dec. 2020*

Graduated Cum Laude (GPA: 3.573 / 4.000)

WORK EXPERIENCE

references available upon request

ENSCO Avionics, Inc.

Embedded Software Engineer II — *Aug. 2022 – May 2023*

- Design tests to verify functionality of the fuel inerting system on the Airbus A321-XLR aircraft
- Utilize HIL testing to verify ICU operation based on the DO-178C verification process
- Utilized DOORS software to write tests to verify software requirements of the inerting system
- Used ARINC interface to verify tests against HIL ICU
- Wrote tests in in-home language to verify software written in C

RESEARCH AND PUBLICATIONS

links available upon request

Risk Aware LTL Motion Planning with Reinforcement Learned Agent and Antagonist

Dr. Dipankar Maity, Ph.D. – University of North Carolina at Charlotte – Charlotte, NC

PROJECTS AND CREATIVE ENDEAVORS

code available on GitHub

NASA University Student Launch Initiative — *2019 – 2020*

First Place Nationally Payload, Second Place Nationally Overall

- Designed, documented, and constructed a rocket (LV) with a quadcopter (UAS) payload
- Lead development of computer systems on UAS and LV
- Lead development of camera vision system to detect ice sample location from LV
- Assisted in design of UAS to retrieve a lunar ice simulant sample
- Assisted in design of deployment system to eject UAS out of LV during decent
- Code was written in C and Python, data was committed to Git source control
- Subscale testing was used to verify instrumentation before flight day
- Various single board computers with I2C, CAN, and SPI communication was utilized to complete challenge

TensorFlow Banana Presence Detector on IoT Arduino board — *Spring 2021*

- In a group, designed, trained, and tested a banana presence detector on a low powered Arduino ARM board
- Would detect presence of banana if placed in front of the camera
- Utilized Arduino Nano 33 BLE board with the OV7670 Camera

IoT Home Security System — *Fall 2020*

- Develop Android app, Raspberry Pi, Web Server, and IoT sensors to remotely control and secure a home
- All applications and server code developed from the ground up using Python, C++, Java, PHP, and SQL
- Embedded IoT sensors communicated with Raspberry Pi dashboard and synced up with an API Server
- Android app controlled the activation of sensors and break-in alarm

SKILLS

Leadership | Troubleshooting | Problem Solving | Communication | Reverse Engineering

Computer Networking | Multithreading | Embedded Systems | Assembly | Software Systems

C++ | Python | Linux | Ansible | Git | OpenCV | Arch Linux | ARM | Java

EXTRACURRICULAR

Boy Scout, Boy Scout of America — *2008 – 2021*

- Assistant Scoutmaster — *Dec. 2019 – Apr. 2021*
- Eagle Scout — *Feb. 2016*

IEEE Eta Kappa Nu (IEEE-HKN) Chapter Member — *Apr. 2022*

NASA USLI Alumni Advisor — 2020 – 2025